GROUP 11 · BLUE OCEAN PITCH

# HYPHA

## Packaging that grows to fit.


Presented by

Matthew Carlin

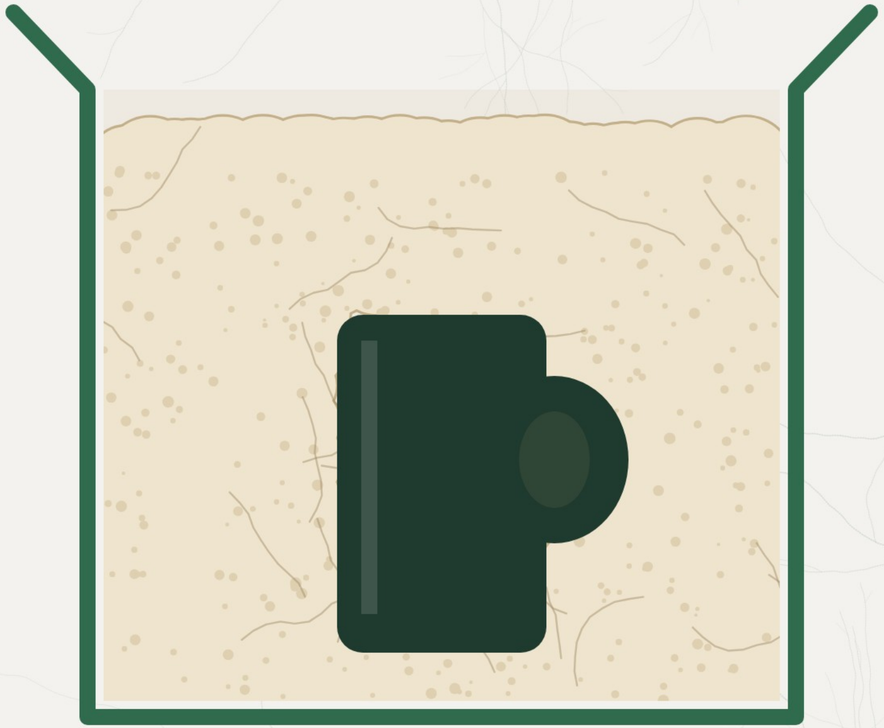
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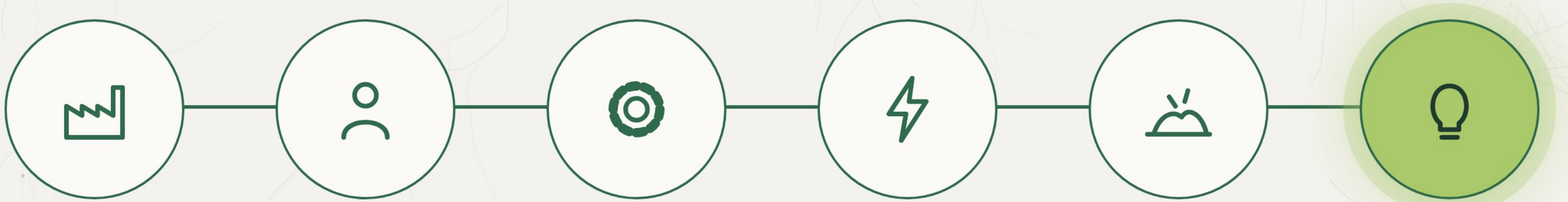



SET · READY TO SHIP

HYPHA PASTE

BEFORE WE BUILT ANYTHING

# We stayed in the problem first.



**Industry**
E-commerce shipping

**People**
Pack stations, 3PLs

**Process**
Pillows, peanuts, paper

**Friction**
Generic fit, overfill

**Consequence**
Damage, landfill, new laws

**Then: ideas**
Only after the evidence

- Evidence before ideas
- Map the whole space
- Ask where it breaks

MIND MAP

# The problem space of packaging

**Packaging**

## Materials

**<10%**




of plastic packaging is **actually recycled**

## Cost

**+28%**




**green premium** for recycled plastic (+20% for paper)

## Waste

**434.5M t**



of packaging waste a year, **projected**

## Manufacturing

**$50K+**




for **one custom mold**, 12 to 24 weeks to make

## Customer

**73%**



say **premium packaging** means a better brand

Sources: Materials [5][6][7] · Cost [10] · Waste [9] · Customer [11] · Mold cost and lead time from team research. Full evidence in the speaker notes; references on the last pages.

THE PROBLEM

# Today, the empty space gets stuffed.

Plastic air pillows | Foam peanuts | Crumpled paper

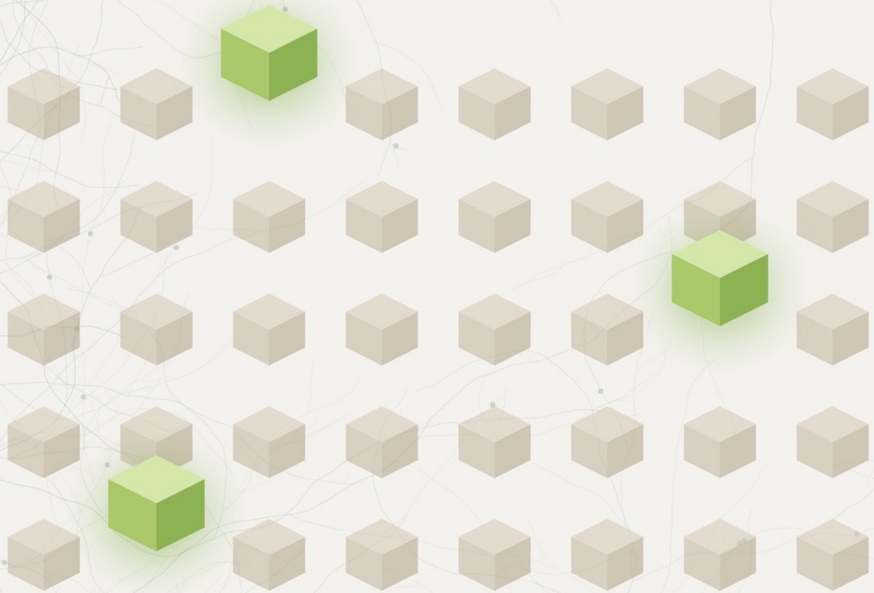
- **Generic fit**
  Products shift and break in transit
- **Rarely recycled**
  Most of it ends up in landfill
- **Under new laws**
  EPR and polystyrene bans target it



Sources: plastic waste and recycling [5][6][7] · EPR and polystyrene bans [1] · Full references on the last pages.

THE PROBLEM · WASTE

# Packaging is used once, then buried.



1 box = 1% of US plastic waste

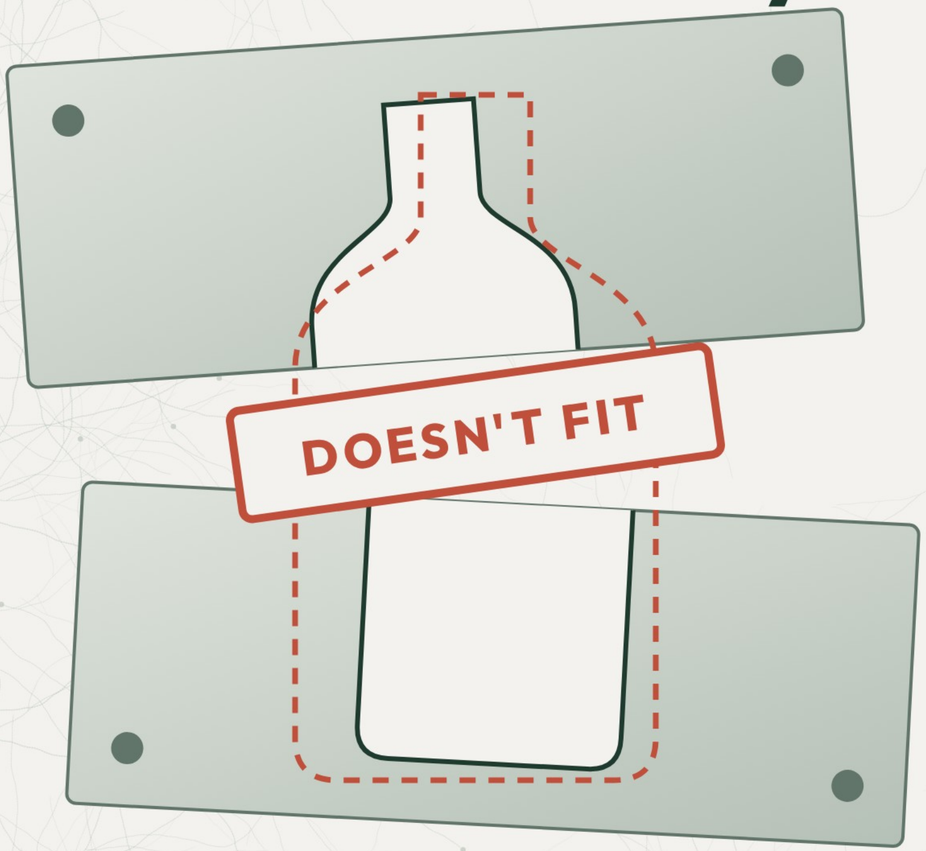
## <10%

of that packaging is recycled.
The rest is buried or burned.

Sources: [5] US EPA, Plastics and Containers & Packaging data · [6] Our World in Data, "Packaging is the source of 40% of the planet's plastic waste" · [7] Beyond Plastics · [8] The Recycling Partnership

THE OTHER OPTION · CUSTOM INSERTS

# A custom fit today means a new mold.




ONE CUSTOM MOLD

## $200K+

starting around $50K

BEFORE THE FIRST PART SHIPS

## 24 weeks

0 — 12 · best case — 24

**×2** Product changes? Pay it all again.

Mold cost and lead time from team research.

THE PROBLEM · CUSTOMER

# The box is the brand's first handshake.

**53%**

came back after a great unboxing



**73%**

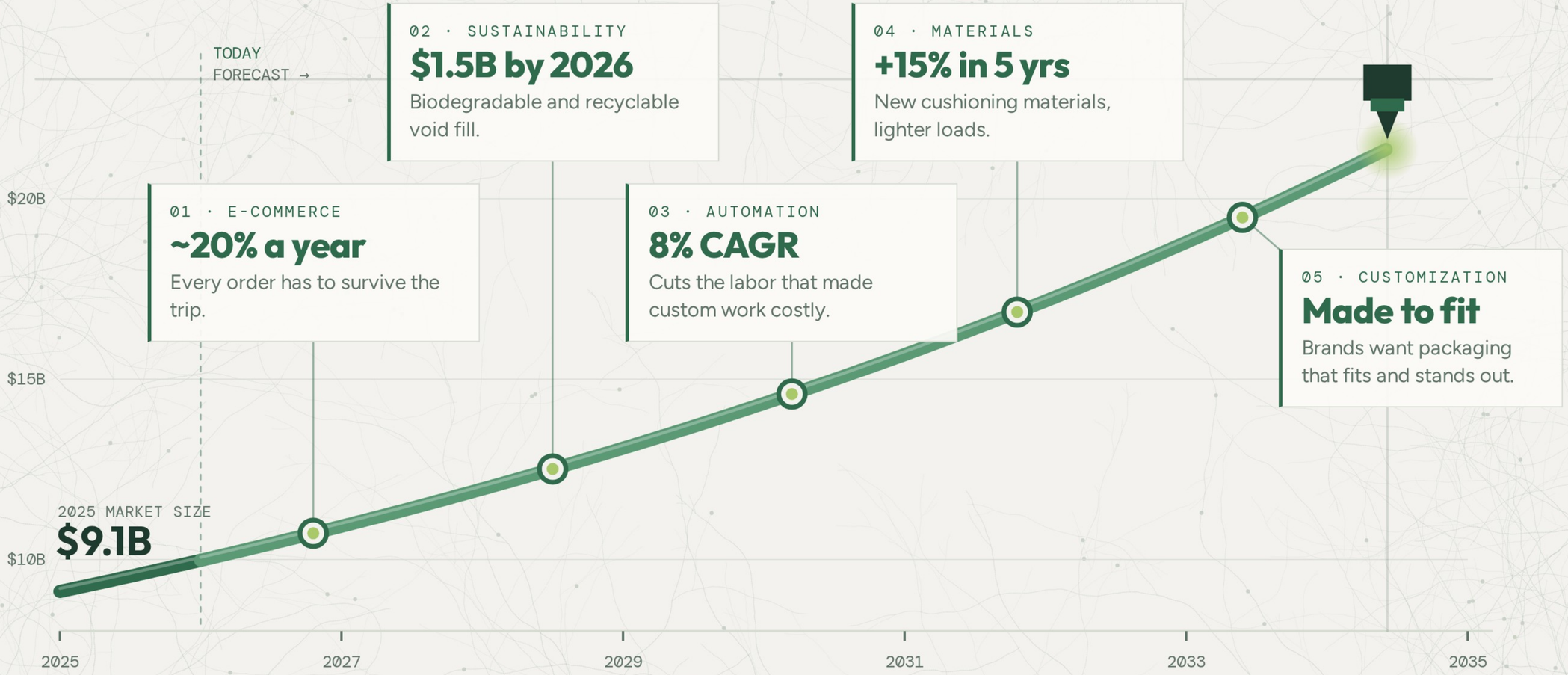
judge the brand by its packaging

**It has to arrive intact.** Protection is why shoppers pick boxes over mailers.

Sources: [13] 2025 study of 1,193 consumers · [14] Journal of Consumer Psychology unboxing experiments · [11] Adobe Express packaging survey · [12] Ryder e-commerce study via Packaging Technology Today

WHY NOW · US & CANADA PROTECTIVE PACKAGING

# A $9.1B market, compounding.



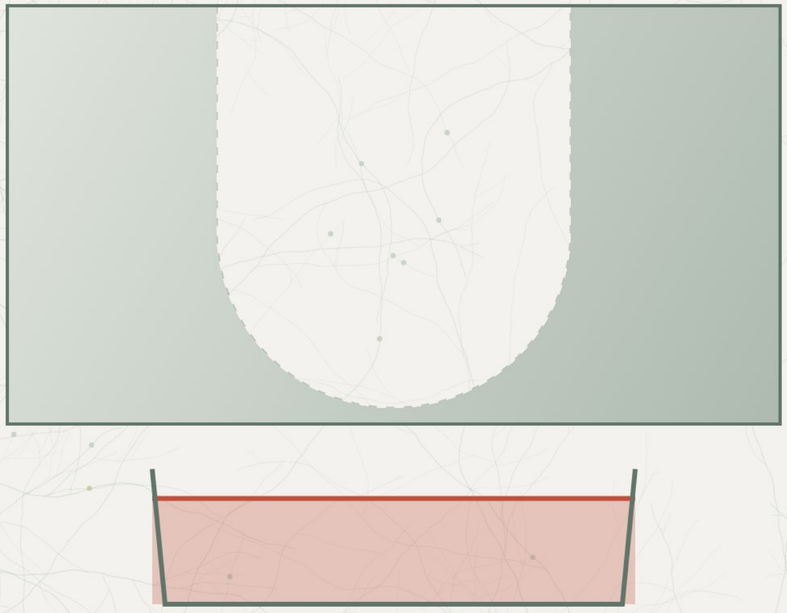

Market size: [1] Future Market Insights, US & Canada Protective Packaging Market (2025, 2035 forecast, 5.1% CAGR as reported). Drivers: [4] Market Research Future, Void Fill Packaging System Market. Curve between the two endpoints is interpolated.

THE EXPONENTIAL DRIVER WE'RE BETTING ON

# Put material only where it's needed.

## Cut it out

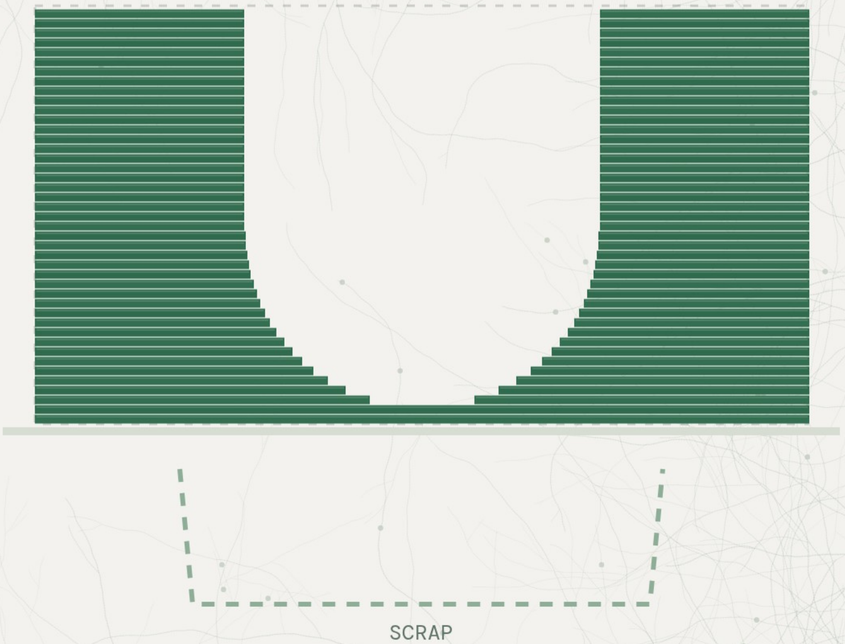
TRADITIONAL




**40%** of the block becomes scrap

## Build it up

ADDITIVE




**0%** scrap, and no mold

Illustration: the scrap share is measured from this drawing, not an industry figure.

PRODUCT DEFINITION · HOW IT WORKS

# Scan. Print. Expand. Return to soil.

A smart void-fill system that 3D-prints biodegradable, mycelium-based foam into the box for a custom fit.

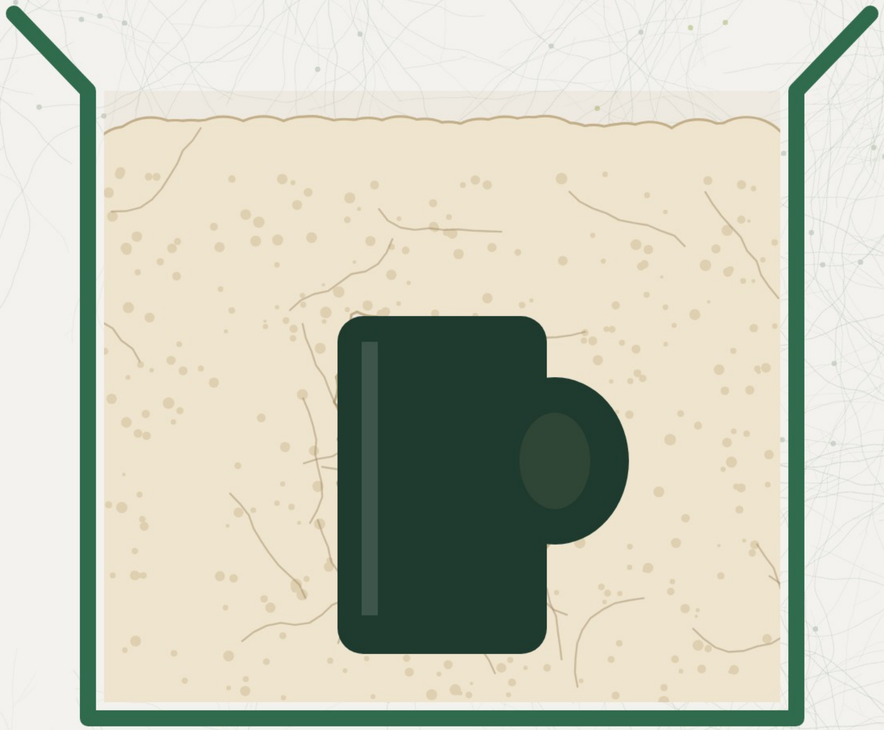
### Scan
A camera maps the box and product in 3D

### Print
A nozzle traces only the empty space

### Expand
The paste foams and locks the product in

### Return to soil
Composts at home in weeks

CONCEPT RENDER



SET · READY TO SHIP

HYPHA PASTE

THE MATERIAL

# Mushroom roots + farm waste.

**Mycelium**
the root network of mushrooms

+

**Hemp hurd**
agricultural waste fiber

+

**Corn husk**
agricultural waste fiber

Home-compostable

Abundant & renewable

Refillable cartridges




REFILLABLE CARTRIDGE

WHO IT'S FOR

# Built for the pack line.

**Fulfillment centers**

**3PLs**

**DTC brands**

Shipping fragile or oddly shaped products: electronics · cosmetics · home goods

### Custom fit, every box

Fewer damaged shipments

### Only what's needed

No overfill

### Plastic-free

Home-compostable

### Drop-in station

Sits at existing pack lines

### A brand moment

Looks engineered, not stuffed

CUSTOMER DISCOVERY · HOLIDAY, DTC APPAREL

# A real brand would switch, up to 2× the cost.

WHAT THEY USE TODAY

**~$1** per order

### Boxes, poly mailers and poly sleeves, bought in bulk

> "Roughly for all shipments, it's about a dollar per order cause we get all the packaging in bulk."

DAMAGE AND UNBOXING

**Very** important

### Damage is rare, but the unboxing is part of the brand

> "We try to make our packaging feel as personal as possible … so the unboxing experience is very important to our brand."

WOULD THEY SWITCH?

**<2×** current cost

### Yes, they'd trial it and possibly switch fully

> "If it surpasses double our current shipping costs, that's when it stops making sense profit-wise."

**Takeaway: Hypha has to land under ~$2 per order, and the unboxing story sells it as much as sustainability does.**

Source: [15] customer interview with the Holiday team, conducted by Ledger (Group 11). One interview; more needed to confirm the price ceiling.

COMPETITIVE ANALYSIS · SWOT

# Where we're strong, and where we're exposed.

## Strengths INTERNAL · HELPFUL

- The only option we found that is both made on demand and home compostable [16]-[22]
- Custom fit for every box with no molds; molded mycelium needs $2K–$9K tooling [20]
- Plastic-free material: mushroom roots plus farm waste

## Weaknesses INTERNAL · HARMFUL

- No prototype yet: expansion, set time and strength are untested
- Cost per order unknown; Holiday pays ~$1 an order and caps a switch at 2× [15]
- Mycelium grows slowly (7 days for molded parts), so the paste must be pre-grown [20]

## Opportunities EXTERNAL · HELPFUL

- US & Canada void fill: $1.9B in 2025, growing to $3.2B by 2035 [1]
- E-commerce is the largest use, at 42.1% of all void fill [2]
- EPR laws (CA, OR, WA) and California's 25% single-use plastic cut by 2032 push brands off plastic [1] [23]

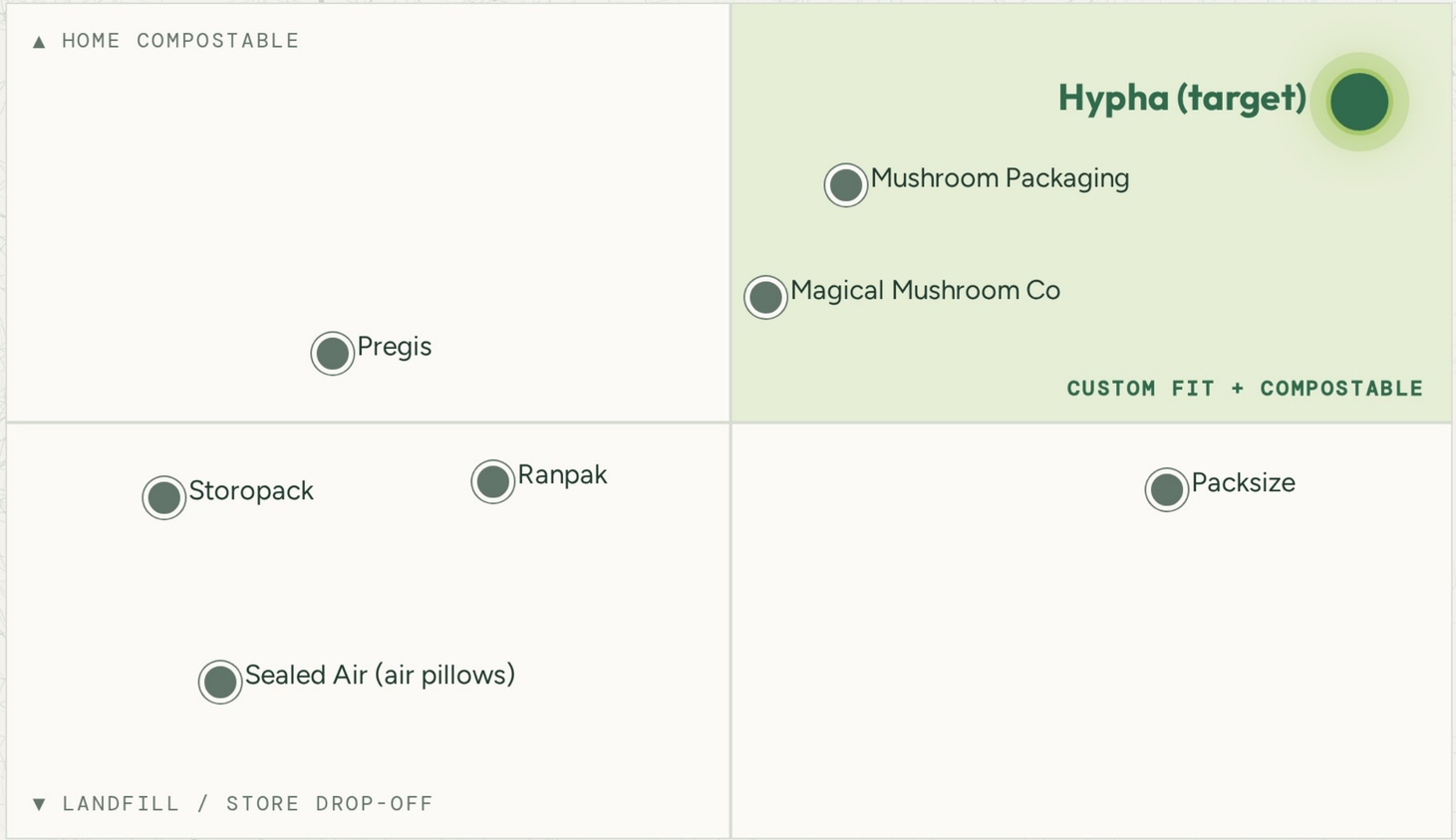
## Threats EXTERNAL · HARMFUL

- Incumbents already run fill machines at the pack line; the top 3 hold 18% of the market [3] [16] [17]
- Ranpak sizes paper fill with AI vision; Packsize's right-sized boxes cut the need for fill [18] [22]
- Paper fill is cheap and curbside recyclable, "green enough" for many buyers [17] [18]

Sources: [1] Future Market Insights, US & Canada Protective Packaging · [2] Market Intelo, Void Fill Packaging · [3] FMI, Void Fill Systems Market Share · [15] Holiday interview (Group 11) · [16] Sealed Air · [17] Pregis · [18] Ranpak · [20] Mushroom Packaging · [22] Packsize · [23] Grand View Research

COMPETITIVE ANALYSIS · LANDSCAPE

# One corner of the map is empty.




READING THE MAP

## Nobody fits each product without plastic or a mold.

01 **On-demand fill is generic.** Sealed Air, Pregis and Ranpak make plastic or paper fill at the pack line.

02 **Green options are molded.** Mycelium is compostable but needs a mold per shape and a 7-day grow.

03 **Hypha takes the open corner:** printed to fit, made on demand, home compostable.

Placed from each company's own product page: [16] Sealed Air · [17] Pregis · [18] Ranpak · [19] Storopack · [20] Mushroom Packaging · [21] Magical Mushroom Co · [22] Packsize. Hypha's spot is a design target. Instapak and Cruz Foam are not placed: no product data in our sources.

COMPETITIVE ANALYSIS · COMPARISON MATRIX

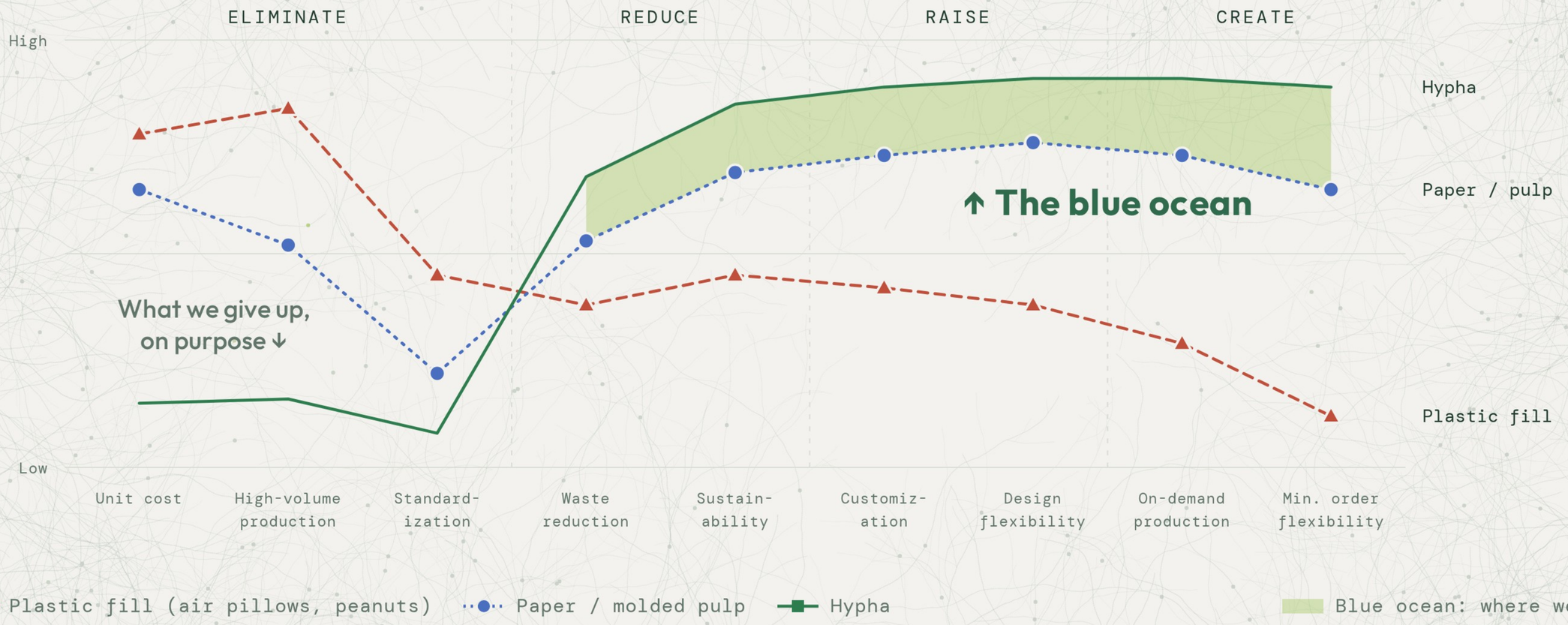
# Side by side, on what buyers care about.

| COMPANY | MATERIAL | ON DEMAND? | FIT | END OF LIFE | MOLDS? | PUBLIC PRICE |
|---|---|---|---|---|---|---|
| **Hypha**<br>Target, not yet built | Mycelium + farm waste | **Yes, at the pack line** | **Custom, every box** | **Home compostable** | **None** | Goal: under $2 per order |
| **Sealed Air**<br>Fill-Air air pillows | Plastic film | **Yes, inflated** | Generic | Store drop-off only | **None** | $835 per roll |
| **Pregis**<br>Paper, air, starch fill | Paper, film, starch | **Yes + pre-made** | Generic | Curbside; Bio One industrial compost | **None** | Not public |
| **Ranpak**<br>FillPak paper systems | Kraft paper | **Yes** | Generic, AI-sized | Curbside recyclable | **None** | Not public |
| **Storopack**<br>PAPERbubble sheets | Paper + air | No, pre-made | Generic | Recyclable | **None** | Not public |
| **Mushroom Packaging**<br>Ecovative | Mycelium + hemp hurd | No, grown ~7 days | **Custom (molded)** | **Home compostable, ~45 days** | $2K–$9K tool | $1,350 prototype fee |
| **Magical Mushroom Co**<br>UK | Mycelium + farm by-products | No, grown | Not stated | Biodegrades ~45 days | Not stated | Claims parity with EPS |
| **Packsize**<br>Right-sized boxes | Corrugated board | **Yes** | Box fits product | Recyclable | **None** | Lease, no list price |

Facts from each company's own page [16]-[22]; company names link to them. $835 per roll is a distributor listing, currency not stated. Green text marks what a buyer wants. The Hypha row is design targets; its price goal comes from our Holiday interview [15].

BLUE OCEAN VALUE MAP

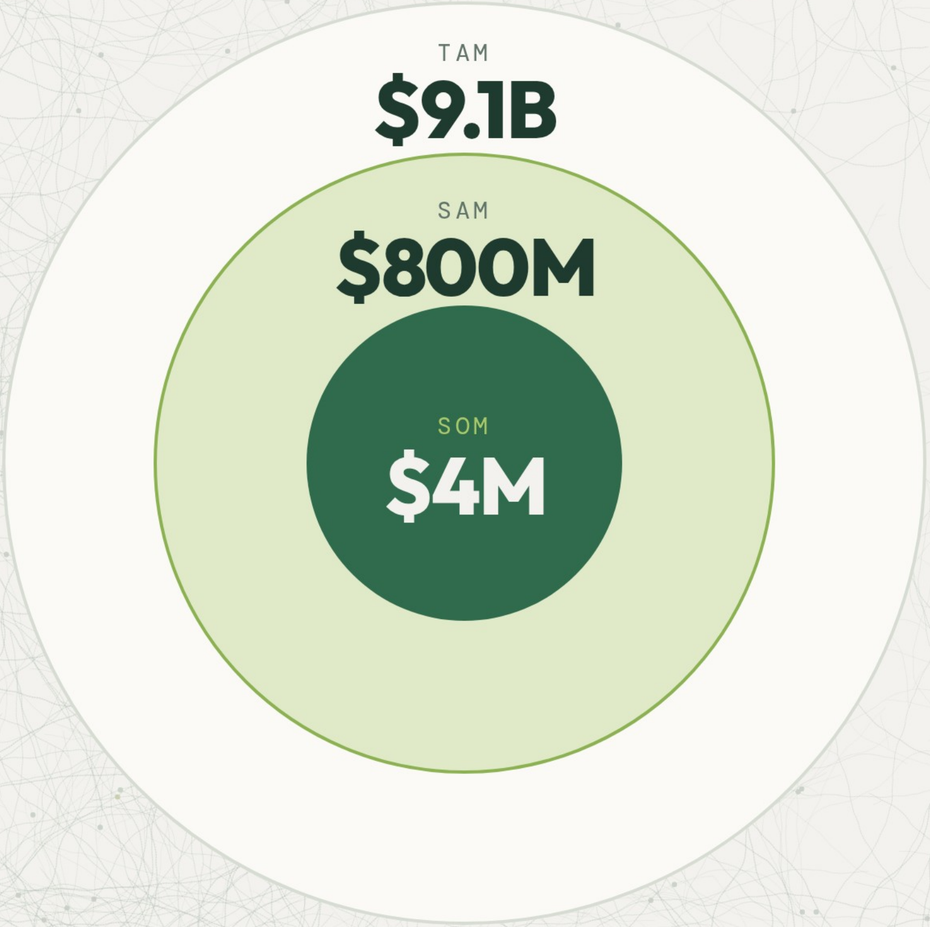
# Stop competing on volume. Compete on fit.




Scores are our team's estimates, to be checked in customer interviews.

MARKET SIZE

# From $9.1B to our first $4M.




TAM

**$9.1B**

## US & Canada protective packaging, 2025

SAM

**$800M**

## E-commerce void fill

`$1.9B void fill` × `42.1% e-commerce` =

`$800M`

SOM

**$4M**

## 0.5% of SAM, benchmarked against today's players

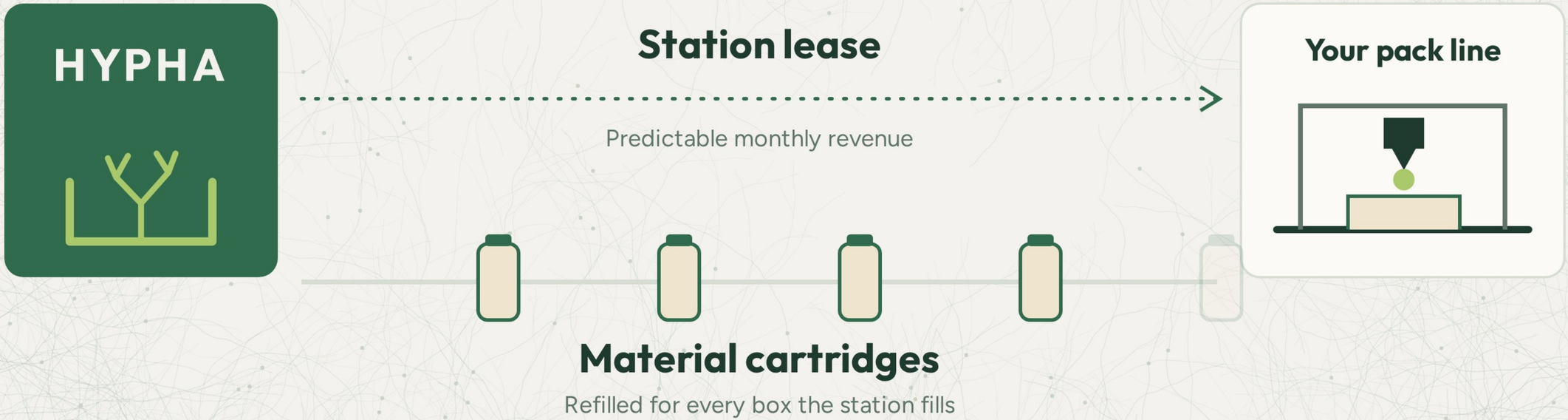
| | |
|---|---|
| Top 10 together | 36% |
| Top 3 together | 18% |
| Each top-3 player | ~6% |
| **Hypha target** | 0.5% |

Conservative case: 0.25% gives about $2M.

Circles not to scale. Sources: [1] Future Market Insights, US & Canada Protective Packaging Market · [2] Market Intelo, Void Fill Packaging Market (e-commerce share) · [3] Future Market Insights, Void Fill Packaging Systems Market Share Analysis. SOM share is our assumption.

BUSINESS MODEL

# Lease the station. Sell the cartridges.




Like a printer and its ink: revenue grows with every box shipped.

VALUE PROPOSITION · GEOFF MOORE TEMPLATE

# Our value proposition, in one sentence.

**For**
TARGET CUSTOMER

direct-to-consumer brands shipping fragile or oddly shaped products

**who**
NEED PROBLEM

want to get plastic out of their packaging without risking broken orders or a cheap-looking unboxing,

**Hypha is**
PRODUCT CATEGORY

an on-demand, compostable void-fill system

**that**
BENEFIT SOLUTION

replaces plastic fill with home-compostable foam made from mushroom roots and farm waste, printed to fit each product so it stays locked in place.

**Unlike**
COMPETITION

air pillows and paper fill, which are the same in every box, or molded mushroom packaging, which needs $2K–$9K tooling and a 7-day grow, Hypha needs no plastic and no molds.

Template: Geoff Moore, *Crossing the Chasm*. Competitor facts: [16] Sealed Air · [18] Ranpak · [20] Mushroom Packaging. Customer need from our Holiday interview [15]. Hypha's benefits are design targets.

VISUAL VALUE PROPOSITION

# All the protection. None of the plastic.

TODAY · GENERIC FILL



The same pillows and peanuts in every box. The product shifts and breaks, and the fill ends up in landfill.

WITH HYPHA · PRINTED TO FIT



Mushroom-based foam printed into only the empty space. The product stays locked in place, then the fill composts at home.

**Custom fit, every box**

**Printed on demand, no molds**

**Mushroom roots + farm waste, back to the soil**

Concept illustration. Custom fit, on-demand printing and home-compostability are Hypha's design targets, not tested results.

WHAT WE BELIEVE

# Packaging shouldn't outlive the order.

Every Hypha fill is built on one belief: **what protects a product for a few days shouldn't sit in a landfill for years.** So we print each fill from mushroom roots and farm waste, **shaped to the product and made to return to the soil.**

REFERENCES

# Sources (1 of 3)


[1] Future Market Insights, Protective Packaging Industry Analysis in United States and Canada, 2025 to 2035 (TAM $9.1B; void fill $1.9B)
https://www.futuremarketinsights.com/reports/united-states-and-canada-protective-packaging-market

[2] Market Intelo, Void Fill Packaging Market Research Report 2034 (e-commerce = 42.1% of void fill, 2025)
https://marketintelo.com/report/void-fill-packaging-market

[3] Future Market Insights, Void Fill Packaging Systems Market Share Analysis (top 3 = 18%, top 10 = 36%)
https://www.futuremarketinsights.com/reports/void-fill-packaging-systems-market-share-analysis

[4] Market Research Future, Void Fill Packaging System Market (growth drivers)
https://www.marketresearchfuture.com/reports/void-fill-packaging-system-market-38467

[5] US EPA, Facts and Figures about Materials, Waste and Recycling: Plastics; Containers and Packaging
https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/plastics-material-specific-data

https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/containers-and-packaging-product-specific

[6] Our World in Data, "Packaging is the source of 40% of the planet's plastic waste"
https://ourworldindata.org/data-insights/packaging-is-the-source-of-40-of-the-planets-plastic-waste

[7] Beyond Plastics, The Real Truth About the U.S. Plastics Recycling Rate
https://www.beyondplastics.org/publications/us-plastics-recycling-rate

[8] The Recycling Partnership, State of Recycling: The Present and Future of Residential Recycling in the U.S. (2024)
https://recyclingpartnership.org/wp-content/uploads/dlm_uploads/2024/01/Recycling-Partnership-State-of-Recycling-Report-1.12.24.pdf

REFERENCES

# Sources (2 of 3)


[9] Smithers, "Packaging waste outpacing global capacity for sustainable management" (434.5M tonnes, 2025)
https://www.smithers.com/resources/2025/december/packaging-waste-outpacing-global-capacity

[10] Packaging Technology Today, "The Rise of Sustainable Packaging in the U.S.: Trends, Drivers, and the Road Ahead"
https://www.packagingtechtoday.com/materials/the-rise-of-sustainable-packaging-in-the-u-s-trends-drivers-and-the-road-ahead/

[11] Adobe Express, "The power of the package" consumer survey (73% say premium packaging improves view of brand)
https://www.adobe.com/express/learn/blog/power-of-the-package

[12] Packaging Technology Today, "Study Finds E-commerce Packaging Choices Impact Purchasing Decisions" (Ryder study; boxes chosen for protection)
https://www.packagingtechtoday.com/featureds/study-finds-e-commerce-packaging-choices-impact-purchasing-decisions/

[13] 2025 consumer study of 1,193 shoppers (unboxing and repeat purchase)
*Cited by the team; link not yet added.*

[14] Journal of Consumer Psychology, unboxing experiments
*Cited by the team; link not yet added.*

[15] Customer interview: Holiday (DTC apparel brand) team, conducted by Ledger, Group 11. Primary research.
*Interview notes held by the team.*

[16] Sealed Air, Fill-Air Extreme inflatable void fill (NASSCO distributor listing)
https://www.nasscoinc.com/item/100699531/Sealed-Air/

REFERENCES

# Sources (3 of 3)


[17] Pregis, Void Fill Packaging (EasyPack paper, AirSpeed air cushions, Bio One loosefill)
https://uk.pregis.com/eu-solutions/by-need/inside-the-box-protection/void-fill-packaging/

[18] Ranpak, Void Fill (FillPak paper systems, DecisionTower AI-sized fill)
https://www.ranpak.com/void-fill/

[19] Storopack, PAPERbubble paper padding
https://www.storopack.us/products/flexible-protective-packaging/paper-padding/paperbubbler/

[20] Mushroom Packaging, a division of Ecovative Design (7-day grow, $2K-$9K tooling, $1,350 prototyping, home compostable)
https://mushroompackaging.com/

[21] Magical Mushroom Company (mycelium packaging, UK)
https://magicalmushroom.com/

[22] Packsize, FAQ (on-demand right-sized boxes)
https://www.packsize.com/resources/faq

[23] Grand View Research, Sustainable Packaging Market Report (California single-use plastic law)
https://www.grandviewresearch.com/industry-analysis/sustainable-packaging-market-report


Notes: Calculated figures (21% void fill share, $800M SAM, $4M SOM) are derived from [1] to [3]. The SOM capture rate (0.5%) is a team assumption benchmarked against incumbent shares. Value-map scores and the additive-manufacturing scrap illustration are team estimates.